Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	[TEAM TO COMPLETE: Official team name]
Selected Domain	Domain 1 — Aiversecity Future-Ready Workforce
Selected SDG Track (1–17)	SDG 8 — Decent Work and Economic Growth (proposed primary); SDG 4 — Quality Education (proposed secondary). [TEAM TO CONFIRM track]
Project / Solution Name	SKILLAX AI Platform
Name (Leader)	[TEAM TO COMPLETE: Team leader's full name]
Email & Phone (Leader)	[TEAM TO COMPLETE: Email] / [TEAM TO COMPLETE: Phone]
University Affiliation(s)	[TEAM TO CONFIRM: Institution(s) for all registered members]
Number of Team Members	[TEAM TO COMPLETE: Number of registered members (Handbook limit: 1–5)]

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is changing job skills. WEF's Future of Jobs 2025 (7 January 2025) estimates 39% of workers' core skills will change by 2030. TalentCorp's Malaysia study (18 November 2024) examined 949 roles in 10 sectors and projects 620,000 employees (18% of those sectors) may be affected. These are projections, not individual skills-gap data. Workers and career changers need clearer ways to compare skills with target roles and choose training. SKILLAX supports that planning; outcomes are unmeasured. Sources: weforum.org/publications/the-future-of-jobs-report-2025 ; talentcorp.com.my/impact-study.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SKILLAX is a career exploration and reskilling prototype for workers, graduates and career changers, with a synthetic-cohort view for workforce planners. Users enter skills or try fictional CVs. Optional Gemini extraction converts CV text into a proposal for user review; an offline keyword fallback is available. Deterministic code compares profiles with ten roles, identifies gaps, models adoption scenarios and builds a four-phase roadmap. Gemini does not rank careers. Hugging Face, MIT News and Crossref provide research-publication activity, not vacancies; workforce examples are synthetic or imported. Workspace data use browser localStorage; no database is configured. Replit configuration exists, but public deployment is unverified. Phase 2 should test users and learning outcomes.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: Next.js 14 App Router, React 18, TypeScript and Tailwind CSS. Backend: Next.js route handlers for CV analysis and research feeds. Gemini uses the official @google/genai SDK for optional CV extraction only; the configured default model is gemini-3.5-flash-lite. Deterministic scoring, skill gaps, scenarios and roadmaps run in TypeScript. Workspace state uses React context and browser localStorage; there is no database. Static catalogs and synthetic workforce observations are bundled in the app. The research route fetches Hugging Face and MIT AI News RSS feeds and Crossref scholarly metadata. Replit configuration exists, but hosting, a public URL and sponsor-provided resources are unverified. No official SDC API or cloud resource is confirmed. See the architecture description in the technical declaration.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The implemented interface includes a profile editor, five-step assessment, synthetic sample profiles, score explanations, skill-gap views, scenario controls and user-managed roadmap milestones. CV extraction is staged for human review. Project records report domain tests and a prior build; these are software checks, not user research. No interviews, usability study, pilot or learning-outcome evaluation is evidenced. The prototype runs locally and includes Replit configuration, but public deployment and judge access are unverified. Next steps are team and mentor review, a defined user test with synthetic or consented data, deployment on a team-selected platform, and pre/post measurement of skill gaps before claiming impact.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4.3 (access to technical/vocational learning), 4.4 (relevant skills for employment), 8.2 (productivity and innovation), and 8.5 (productive employment). SKILLAX supports career exploration and planning; it has no measured SDG contribution. UN: https://sdgs.un.org/goals/goal4 and https://sdgs.un.org/goals/goal8 (accessed 24 Oct 2026).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	TalentCorp's Malaysia Impact Study examined 949 roles in 10 sectors and estimates about 620,000 employees (about 18% of the analysed-sector workforce) may be significantly affected. This projected exposure is not a measured SKILLAX-user skills gap. No user-cohort gap baseline exists. Source: https://www.talentcorp.com.my/impact-study/ (launched 18 Nov 2024).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Mean role-specific skill-gap level per profile (0–5): sum positive differences between role-required and user-rated skill levels divided by number of listed role skills. Report cohort size and completion rate; self-ratings are not independent competency validation.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed, not achieved: target at least 20% reduction in the cohort mean skill-gap score after a six-week guided-roadmap pilot, using the same rubric before and after. Define the cohort and protocol; report attrition and use a comparison group if feasible. [TEAM TO APPROVE]

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☐ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☐ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☐ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☒ Yes ☐ No [TEAM TO CONFIRM: identify pre-existing work, owners and dates.]

If yes, identify pre-existing SKILLAX code/design/research, dates and owners: [TEAM TO COMPLETE].

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js 14.2.35; React 18.3.1; Zod; Tailwind CSS	App, UI, validation and styling	MIT
@google/genai 2.31.0; Gemini	CV extraction SDK and hosted model	Apache-2.0; Google API terms
Synthetic profiles and workforce data	Runtime demo inputs; not training data	Project-authored; synthetic

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No [TEAM TO CONFIRM: review each member's employment, internship and consulting agreements.]

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[TEAM LEADER TO SIGN]

[TEAM TO COMPLETE: Date of signing]